

Autodesk® **Scaleform®**

Scaleform

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Autodesk® Scaleform® 4.3

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 - 2.2.1 内存 存

1 Scaleform 3.x

Autodesk® Scaleform® 3.0
Scaleform 3.3 (AMP) Scaleform 3.2
1.2

Scaleform 3.3

-
- MovieViews Scaleform
- ActionScript
-

1.1

Scaleform
/ Scaleform

1.1.1 (Override Allocator)

Scaleform Gfx::System

1. Scaleform::SysAlloc Alloc/Free/Realloc
dlmalloc

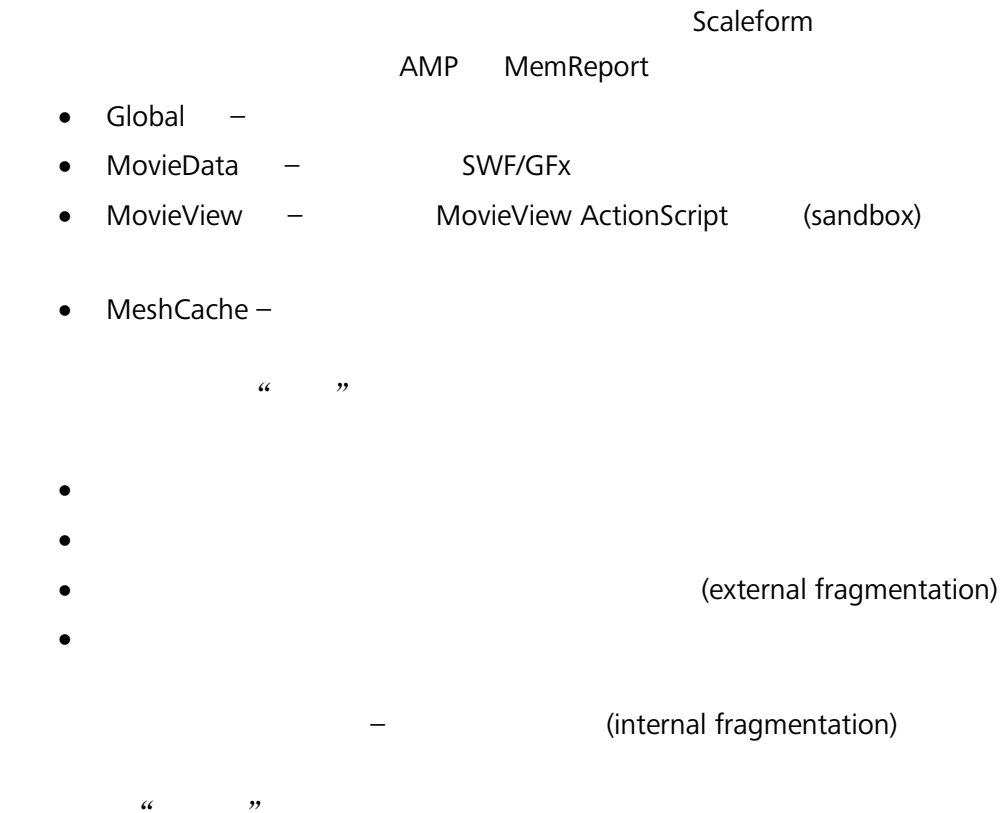
2. Scaleform
“ ” (Pause Menu) (Memory Arena)

3. Scaleform
(Hardware Paging) Scaleform::SysAllocWinAPI

(2)

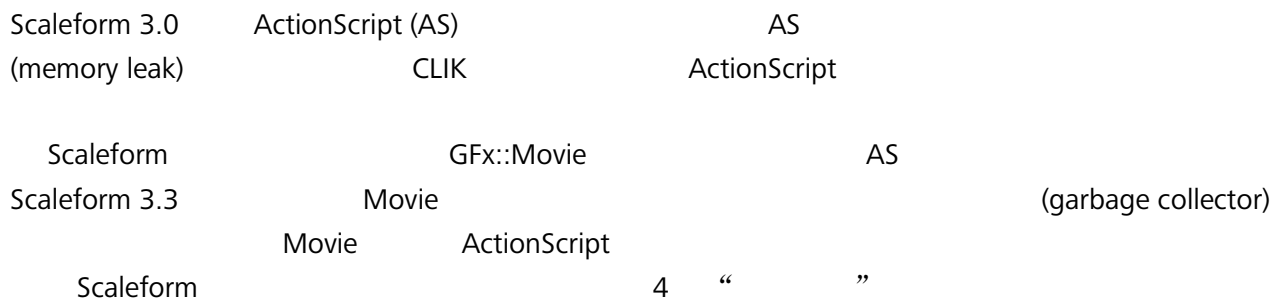
2 “ ”

1.1.2 (Memory Heap)

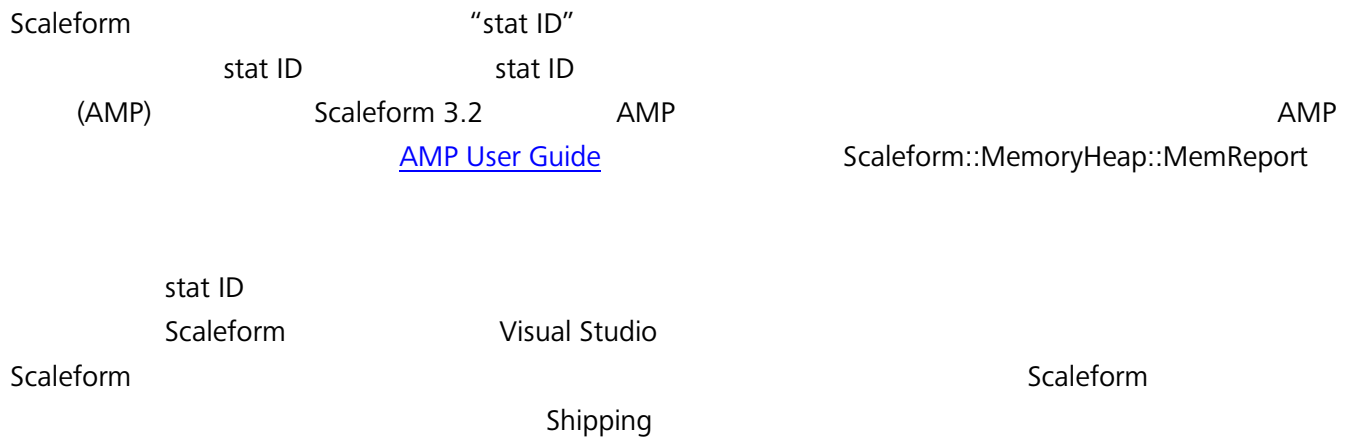


Scaleform 3.3 GSysAlloc
 Scaleform 4.0 GFx::Movie

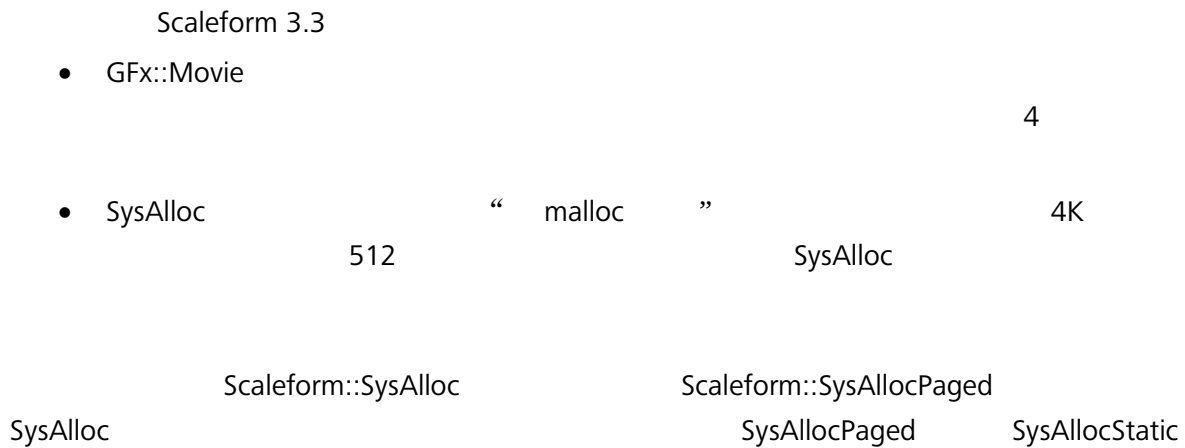
1.1.3 (Garbage Collection)



1.1.4



1.2 Scaleform 3.3 API



2

1.1.1 Scaleform

1. SysAlloc
2. Scaleform
3. Scaleform

(1)

(2)

SysAlloc Scaleform Scaleform
SWF Scaleform Scaleform::SysAlloc::Alloc
 Scaleform::SysAlloc::Free Scaleform

 (compartmentalization)
2.1 SysAlloc

(level loading)

2.2

“ ”

Scaleform
Scaleform

—

2.1

Scaleform

1. SysAlloc GetInfo Alloc Free
2. Scaleform Gfx::System

Scaleform

malloc/free

Scaleform::SysAllocMalloc
(alignment)

2.1.1 SysAlloc

Scaleform SysAllocMalloc

SysAllocMalloc

```
class MySysAlloc : public SysAlloc
{
public:
    virtual void* Alloc(UPInt size, UPInt align)
    {
        return _aligned_malloc(size, align);
    }

    virtual void Free(void* ptr, UPInt size, UPInt align)
    {
        SF_UNUSED2(size, align);
        _aligned_free(ptr);
        return true;
    }

    virtual void* Realloc(void* oldPtr, UPInt oldSize,
                          UPInt newSize, UPInt align)
    {
        SF_UNUSED(oldSize);
        return _aligned_realloc(oldPtr, newSize, align);
    }
};
```

			MySysAlloc	SysAlloc	
GetInfo	Alloc	Free			Scaleform
	16	oldSize	align	Free/Realloc	
		Alloc/Free		Realloc	

Scaleform

Gfx::System

```
MySysAlloc myAlloc;
Gfx::System gfxSystem(&myAlloc);
```

Scaleform 3.0	Gfx::System	Scaleform	Scaleform
			Scaleform
		Scaleform	Gfx::System
		Gfx::System::Init()	Gfx::System::Destroy()

Gfx::System

Gfx::System::Init()

SysAlloc

2.2

Scaleform

Scaleform

SysAlloc

SysAllocStatic

```
void*
SysAllocStatic
Gfx::System
...
pmemChunk = ...;
blockAlloc(pmemChunk, 6*1024*1024);
gfxSystem(&blockAlloc);
```

6

Scaleform

gfxSystem

blockAlloc

“ ”

Scaleform

SysAllocStatic

Alloc

0

Scaleform

SysAllocPaged

2.2.1

Scaleform 3.1

Scaleform

Gfx::Movie

Scaleform

Scaleform

“ ”

2.2.1.1

Scaleform

Scaleform

Gfx::System

SysAlloc

Scaleform

CreateMovie

Scaleform

Scaleform

Scaleform

SDK

Scaleform

```

pMovieDef->WaitForLoadFinish(true);
pMovie      = 0;
pMovieDef = 0;

```

or (if Scaleform::Ptr is not used):

```

pMovie->Release();
pMovieDef->Release();

```

5. DestroyArena

CreateArena Memory::ArenalsEmpty

```

//                               DestroyArena
//                               "return *(int*)0;"
Memory::DestroyArena(1);

```

SysAlloc “ pbuff1 ”

2.3

OS

- *Scaleform::SysAllocWinAPI* – Scaleform OS- VirtualAlloc/VirtualFree API
- *Scaleform::SysAllocPS3* – sys_memory_allocate/sys_memory_free PS3 (OS)

Scaleform WinAPI PS3 SysAlloc

“ ”

dlmalloc 2MB

2.3.1 SysAllocPaged

Scaleform 3.3 SysAlloc SysAlloc SysAllocPaged
Scaleform SysAllocStatic OS SysAllocPaged SysAlloc

SysAllocMalloc

```
class MySysAlloc : public SysAllocPaged
{
public:
    virtual void GetInfo(Info* i) const
    {
        i->MinAlign      = 1;
        i->MaxAlign      = 1;
        i->Granularity    = 128*1024;
        i->HasRealloc     = false;
    }

    virtual void* Alloc(UPInt size, UPInt align)
```

```

{
    // Ignore 'align' since reported MaxAlign is 1.
    return malloc(size);
}

virtual bool Free(void* ptr, UInt size, UInt align)
{
    // free() doesn't need size or alignment of the memory block, but
    // you can use it in your implementation if it makes things easier.
    free(ptr);
    return true;
}
};

```

			MySysAlloc	SysAllocPaged
GetInfo	Alloc	Free		ReallocInPlace

1 GetInfo() – Scaleform::SysAllocPaged::Info

2 Alloc – size align Alloc
align GetInfo MaxAlign MaxAlign 1
align

3 Free – Alloc size align

4 ReallocInPlace – false

[Scaleform 参考资料](#)

	Alloc	Free		GetInfo
Scaleform	Scaleform			SysAllocPaged
			SysAllocPaged::Info	

- MinAlign –
- MaxAlign – 0 Scaleform
1 Alloc align
- Granularity – Scaleform Scaleform
malloc 64K
- HasRealloc – ReallocInPlace

- SysDirectThreshold
SysDirectThreshold
- MaxHeapGranularity
MaxHeapGranularity alloc/free
4096 4096

MaxAlign SysAlloc

1. MaxAlign 1 Scaleform
2. MaxAlign 0
3. Alloc MinAlign MaxAlign

Scaleform GFx::System

```
MySysAlloc myAlloc;
GFx::System gfxSystem(&myAlloc);
```

2.4

Scaleform “ ”
(Mesh Cache) Scaleform
API

2.4.1 API

Scaleform
“ new ”

```
Ptr<GFx::FileOpener> opener = *new GFx::FileOpener();
loader.SetFileOpener(opener);
```

‘ new ’ Scaleform
GFx::System GFx::System
0

```

Gfx::MeshCacheManager
Gfx::Loader::CreateMovie()
Gfx::MovieDef::CreateInstance
SWF/GFX
ActionScript
Scaleform

```

```

Scaleform::MemoryHeap
Scaleform::MemoryHeap::CreateHeap
Scaleform::MemoryHeap::Release
Scaleform::MemoryHeap::Free
SysAlloc
Scaleform
Scaleform::MemoryHeap

```

- SF_HEAP_ALLOC(heap, size, id)
- SF_HEAP_MEMALIGN(heap, size, alignment, id)
- SF_HEAP_NEW(heap) ClassName
- SF_FREE(ptr)

‘ id ’

Scaleform AMP

```

Gfx::SpriteDef
‘ pHeap ’

```

```

Ptr<Gfx::SpriteDef> def = *SF_HEAP_NEW(pHeap) Gfx::SpriteDef(pdataDef);

```

```

Gfx::SpriteDef
Scaleform
Scaleform
Ptr<>
Scaleform::NewOverrideBase
‘ delete ’
delete
SF_FREE

```


2.4.1

Scaleform::String

Scaleform::Array<>

```
class Sprite : public RefCountBase<Sprite>
{
    String          Name;
    Array<Ptr<Sprite> > Children;

    Sprite(String& name, ...) { ... }
};
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);
sprite->DoSomething();
```

GFx::Sprite

-
GFx::Sprite

Scaleform

“ ”

- SF_HEAP_AUTO_ALLOC(ptr, size)
- SF_HEAP_AUTO_NEW(ptr) ClassName

SF_HEAP_ALLOC SF_HEAP_NEW

```
MemoryHeap*    pheap = ...;
UByte *        pBuffer = (UByte*)SF_HEAP_ALLOC(pheap, 100,
GFx::StatMV_ActionScript_Mem);
Ptr<GFx::Sprite> sprite = *SF_HEAP_AUTO_NEW(pBuffer + 5) GFx::Sprite();
...
sprite->DoSomething();
...
// Release sprite and free buffer memory.
sprite = 0;
SF_FREE(pBuffer);
```

```

Gfx::Sprite
        ' pbuffer '          SF_HEAP_AUTO_NEW
Scaleform

```

Gfx::Sprite

```

class Sprite : public RefCountBase<Sprite>
{
    StringLH          Name;
    ArrayLH<Ptr<Sprite> > Children;

    Sprite(String& name, ...) { ... }
};
Ptr<Sprite> sprite = *SF_HEAP_NEW(pHeap) Sprite(name);
sprite->DoSomething();

```

```

Scaleform::String  Scaleform::Array<>          "      "
Scaleform::StringLH  Scaleform::ArrayLH<>
    "      "
    Scaleform

```

2.4.2

```

GfxConfig.h          SF_MEMORY_TRACE_ALL
Scaleform::Memory     SetTracer          CreateHeap DestroyHeap Alloc Free

```

```

SysAllocWinAPI          :

#include "Kernel/HeapPT/HeapPT_SysAllocWinAPI.h"

class MyTracer : public Scaleform::MemoryHeap::HeapTracer
{
public:
    virtual void OnCreateHeap(const MemoryHeap* heap)
    {
        //...
    }

    virtual void OnDestroyHeap(const MemoryHeap* heap)
    {
        //...
    }
}

```

```

    }

    virtual void OnAlloc(const MemoryHeap* heap, UPInt size, UPInt align,
                        unsigned sid, const void* ptr)
    {
        //...
    }

    virtual void OnRealloc(const MemoryHeap* heap, const void* oldPtr,
                        UPInt newSize, const void* newPtr)
    {
        //...
    }

    virtual void OnFree(const MemoryHeap* heap, const void* ptr)
    {
        //...
    }
};

static MyTracer tracer;
static SysAllocWinAPI sysAlloc;
static Gfx::System* gfxSystem;

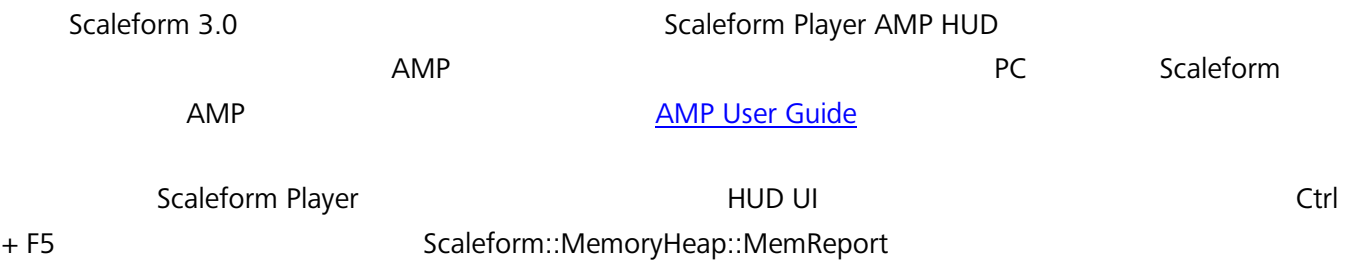
class MySystem : public Scaleform::System
{
public:

    MySystem() : Scaleform::System(&sysAlloc)
    {
        Scaleform::Memory::GetGlobalHeap()->SetTracer(&tracer);
    }
    ~MySystem() { }
};

SF_PLATFORM_SYSTEM_APP(FxPlayer, MySystem, FxPlayerApp)

```

3



```
void MemReport (class StringBuffer& buffer, MemReportType detailed,
                bool xmlFormat = false);
void MemReport (struct MemItem* rootItem, MemReportType detailed);
```

MemReport 生成一份内存报告，可能的格式是 XML，写到一个提供的字符串缓冲区。

MemReportType

- MemoryHeap::MemReportBrief – Scaleform

Memory 4,680K / 4,737K	
Image	1,052
Sound	136
Movie View	1,739,784
Movie Data	300,156

- MemoryHeap::

[Heap] Global	4,848,864
Heap Summary	
Total Footprint	4,848,864
Local Footprint	2,427,860
Child Footprint	2,421,004
Child Heaps	4
Local Used Space	2,417,196
DebugInfo	120,832
Memory	
MovieDef	
Sounds	8
ActionOps	80,476
MD_Other	200
MovieDef	36
MovieView	
MV_Other	32
General	91,486
Image	92
Sound	136
String	31,425
Debug Memory	
StatBag	16,384
Renderer	
Buffers	2,097,152
RenderBatch	5,696
Primitive	1,512
Fill	900
Mesh	4,884
MeshBatch	2,200
Context	15,184
NodeData	15,160
TreeCache	13,220
TextureManager	2,336
MatrixPool	4,096
MatrixPoolHandles	2,032
Text	1,232
Renderer	13,488
Allocations Count	1,822

- MemoryHeap:: MemReportFileSummary –

stat ID	SWF	MovieData	MovieDef	MovieView
Movie File	3DGenerator_AS3.swf			
Memory				1,956,832
MovieDef				152,664

CharDefs	39,848
ShapeData	1,716
Tags	73,656
MD_Other	37,444
MovieView	196,249
MovieClip	120
ActionScript	181,917
MV_Other	14,212
General	1,596,712
Image	960
String	2,783
Renderer	7,464
Text	7,464

- MemoryHeap::MemReportHeapDetailed -

AMP

Scaleform

Total Footprint	4,858,148
Used Space	4,793,972
Global Heap	2,405,492
MovieDef	80,720
Sounds	8
ActionOps	80,476
MD_Other	200
MovieView	32
MV_Other	32
General	92,566
Image	92
Sound	136
String	31,276
Debug Memory	8,192
StatBag	8,192
Renderer	2,182,960
Buffers	2,097,152
RenderBatch	5,696
Primitive	1,512
Fill	900
Mesh	4,884
MeshBatch	2,200
Context	15,184
NodeData	15,016
TreeCache	13,136
TextureManager	2,336
MatrixPool	8,192
MatrixPoolHandles	2,032
Text	1,232
Movie Data Heaps	300,156

MovieData "3DGenerator_AS3.swf"	300,156
MovieDef	152,664
CharDefs	39,848
ShapeData	1,716
Tags	73,656
MD_Other	37,444
General	141,332
Image	960
String	2,312
Movie View Heaps	1,727,936
MovieView "3DGenerator_AS3.swf"	1,727,936
MovieView	195,977
MovieClip	120
ActionScript	181,645
MV_Other	14,212
General	1,467,668
String	471
Renderer	7,464
Text	7,464
Other Heaps	360,580
_FMOD_Heap	360,580
General	359,944
Debug Data	12,884
Unused Space	51,292
Global	51,292
_FMOD_Heap	4,384
MovieData "3DGenerator_AS3.swf"	5,356
MovieView "3DGenerator_AS3.swf"	33,032

SF_MEMORY_ENABLE_DEBUG_INFO Scaleform stat ID
stat ID

- MovieDef –
- MovieView – Gfx::Movie ActionScript
- CharDefs – Flash
- ShapeData –
- Tags –
- ActionOps – ActionScript
- MeshCache – (vector tessellator) EdgeAA
- FontCache –

4

Scaleform 2.2

ActionScript

ActionScript

“circular references”

“cycle references”

Code:

```
var o1 = new Object;  
var o2 = new Object;  
o1.a = o2;  
o2.a = o1;
```

ActionScript

Flash UI

Scaleform 3.0

-

“

” Scaleform

ActionScript

“ ”

Flash

Scaleform 3.x

Scaleform

Include/GFxConfig.h

GFX_AS_ENABLE_GC

```
// Enable garbage collection  
#define GFX_AS_ENABLE_GC
```

Scaleform

4.1

Scaleform 3.x

MovieView

MovieDef

```
Gfx::Movie* Gfx::MovieDef::CreateInstance(const MemoryParams& memParams,
                                          bool initFirstFrame = true)
```

MemoryParams

Gfx::MovieDef

```
struct MemoryParams
{
    MemoryHeap::HeapDesc Desc;
    float HeapLimitMultiplier;
    unsigned MaxCollectionRoots;
    unsigned FramesBetweenCollections;
};
```

- Desc – (Granularity) (Limit) (Flags) MinAlign Scaleform
- HeapLimitMultiplier – 0...1
- MaxCollectionRoots – roots roots
- FramesBetweenCollections – roots

Movie View

```
MemoryContext* MovieDef::CreateMemoryContext(const char* heapName,
                                              const MemoryParams& memParams,
                                              bool debugHeap )
```

```
Movie* MovieDef::CreateInstance(MemoryContext* memContext,
                                bool initFirstFrame = true)
```

MemoryContext	Movie View		
Movie View		AMP	
		AMP	Movie View

```

MemoryParams::Desc
ActionScript
    128K(
        Desc.Limit  HeapLimitMultiplier
        Boehm-Demers-Weiser (BDW)
    )

```

BDW

```

    if (allocs since collect >= heap footprint * HeapLimitMultiplier)
        collect
    else
        expand(heap footprint + overlimit + heap footprint *
            HeapLimitMultiplier)

```

“ collect ” ActionScript

HeapLimitMultiplier	0.25	25% HeapLimitMultiplier
Scaleform		25%

Desc.Limit

Desc.Limit

	MaxCollectionRoots/FramesBetweenCollections
MaxCollectionRoots/	FramesBetweenCollections.MaxCollectionRoots roots

“ root ”	ActionScript	ActionScript
----------	--------------	--------------

ActionScript	ActionScript	ActionScript
--------------	--------------	--------------

ActionScript	“ obj.member = 1 ”	“ obj ”	roots
ActionScript	root	MaxCollectionRoots	ActionScript
		MaxCollectionRoots	

0

FramesBetweenCollections

“frame ”	Flash	SWF	30 fps	30	2
----------	-------	-----	--------	----	---

60	Advance		
	1800	60	30 fps
FramesBetweenCollections		Desc.Limit	ActionScript
	0		

4.2 Gfx::Movie:: ForceCollectGarbage

```
virtual void ForceCollectGarbage() = 0;
```

off